PROPOSED BUDGET

Fungal and microbial colonisation of tropical educational building exteriors: molecular characterisation and campus air-quality context at UTS Sibu

Dr Armstrong Ighodalo Omoregie | BERI, School of Built Environment, UTS | UTS Matching Grant, Cycle 1/2027 | Year 1: Mar 2027 to Feb 2028, Year 2: Mar 2028 to Feb 2029

VOT	Category	Justification and detailed breakdown (Year 1 and Year 2)	Year 1 (RM)	Year 2 (RM)	Total (RM)
11000	**Salary and Wages**	**Y1 (RM1,500):** Undergraduate research assistant allowance at RM7.00/hour, max 4 hours/day (RM28.00/day), approx. 18 working days per active month = RM500.00/month for 3 active months, timed to sampling rounds R1-R3 and monitoring node assembly. **Y2 (RM1,500):** Same rate, 3 active months, for rounds R4-R6, plate inoculation and cold-chain packing. Each round requires swabbing 36 fixed points across 4 buildings and 4 facade orientations, on-site inoculation of agar plates, recording of covariates and algal cover, and packing for transfer to SRDC-TROPI: approx. 2 field days plus 1 preparation day per round. The allocation purchases approx. 107 assistant-days. Within the RM3,000 ceiling for temporary and contract personnel, Guideline Section 7(a).	1,500.00	1,500.00	3,000.00
21000	**Travel and Transport**	**Y1 (RM850):** Cold-chain transport of biological materials, UTS Sibu to SRDC-TROPI Kota Samarahan (approx. 300 km): 2 inbound consignments of prepared swabs and media and 3 return consignments of inoculated plates, 5 legs at RM150.00 = RM750.00; inbound delivery of sensor modules, enclosures and field consumables, 2 deliveries at RM50.00 = RM100.00. **Y2 (RM2,150):** 1 inbound and 3 return consignments, 4 legs at RM150.00 = RM600.00; 1 delivery at RM50.00; attendance at one indexed national conference to present project findings (RM1,500.00) comprising return airfare RM700.00, accommodation RM250.00/night x 3 nights = RM750.00, ground transport RM50.00. Media are batched across rounds inbound, while returns follow each round individually because the cultures are live and require sealed triple packaging under cold chain, so ordinary post is not usable. Conference registration is claimed separately under VOT 29000 as the guideline directs. Subsistence claimed only for paper presentation. Local Sibu fieldwork travel excluded, as all four sampled buildings are on the UTS campus. Within the RM3,000 SET ceiling, Guideline Section 7(b).	850.00	2,150.00	3,000.00
24000	**Rental**	**Y1 (RM1,200):** Rental of a reference-grade air quality analyser for co-location calibration of the low-cost monitoring nodes under Sibu field conditions, RM600.00/week x 2 weeks: one pre-deployment campaign and one post-installation verification. Low-cost optical particulate sensors are subject to hygroscopic bias at the high relative humidity prevailing on this campus and metal oxide gas sensors drift; co-located reference measurement is what allows a correction to be derived and the reported values to be defensible. Rental rather than purchase is appropriate for two short campaigns. Placed in VOT 24000 rather than VOT 35000 per Guideline Section 7(c). **Y2 (RM0):** No further rental required.	1,200.00	0.00	1,200.00
26000	**Research Materials and Supplies**	**Y1 (RM9,300):** Monitoring node development (RM8,040) for 4 campus nodes plus spares, in a configuration already validated in deployed service by the co-researcher under existing work, removing design risk: 4 multi-parameter air-quality modules measuring CO2, PM2.5, PM10, temperature and humidity (RM650.00 ea, RM2,600.00); 8 discrete temperature/humidity sensors as redundant per-node reference (RM25.00 ea, RM200.00); 4 metal oxide gas sensor sets for CO, CO2, O3, NH3, NOx (RM280.00/set, RM1,120.00); 6 microcontroller boards with breakout (RM90.00 ea, RM540.00); 1 LoRa gateway and transmitter set (RM900.00); 4 IP65 ventilated enclosures with mounting (RM220.00 ea, RM880.00); 4 sets power supply, solar panel, battery, cabling and connectors (RM300.00/set, RM1,200.00); PCB fabrication, prototyping and soldering consumables (RM600.00). Field sampling consumables (RM1,260): 10 stainless sterile sampling templates 10 x 10 cm (RM600.00), 2 cold-chain coolers with gel packs (RM360.00), PPE and field consumables for R1-R3 (RM300.00). **Y2 (RM2,500):** 2 further coolers (RM360.00); PPE and field consumables for R4-R6 (RM400.00); ethanol, sample bags, labels and markers (RM540.00); culture consumable contingency (RM1,200.00) for SDA and PDA media, sterile swabs and Petri dishes. SRDC-TROPI supplies culture media and swabs from its matched expenditure, so the contingency covers only control plates required by the contamination-control protocol, repeat sampling after a failed batch, and any delay in partner supply; it does not duplicate the partner's provision. Sulphur dioxide sensing excluded, as it bears on no mechanism this project tests. Guideline Section 7(d).	9,300.00	2,500.00	11,800.00
29000	**Professional Services**	**Y1 (RM1,300):** Sanger sequencing of ITS amplicons, 32 reactions at the quoted rate of RM25.00/reaction (RM800.00) for isolates from rounds R1-R3, inclusive of sample QC by gel electrophoresis and one free re-run where a result is unsatisfactory; SORAS application and Sarawak Biodiversity Council permit fees covering the project, its scope, its personnel and the participating institutions (RM500.00). **Y2 (RM1,700):** Sanger sequencing, 48 reactions at RM25.00 (RM1,200.00) for isolates from rounds R4-R6; registration fee for one indexed national conference (RM500.00). The 80 reactions cover a curated set of up to 60 isolates by single-pass read, with bidirectional reads where assignment is ambiguous. DNA extraction and PCR amplification are performed by SRDC-TROPI as an in-kind matched contribution, so only the sequencing service is charged to this grant. Research may not be conducted in Sarawak without a valid permit, so the permit fee is a precondition of all fieldwork. Journal article processing charges excluded; non-APC journals targeted for both indexed outputs. Within the RM3,000 SET ceiling, Guideline Section 7(f).	1,300.00	1,700.00	3,000.00
35000	**Accessories and Small Appliances**	**Y1 (RM3,000):** Outdoor-rated display panel presenting live readings and the current campus air-quality classification at a high-traffic campus location (RM2,200.00); weatherproof housing, mounting bracket and installation hardware (RM800.00). The display is a deliverable of the collaboration agreement between UTS and SRDC and is the pathway by which monitoring becomes visible to building occupants and supports institutional environmental reporting. Cloud hosting and the data plan are excluded, as the guideline does not support utilities or subscriptions under VOT 26000 and VOT 23000 is restricted to USRRG and UFSRRG; these are provided as a UTS in-kind contribution through institutional IT infrastructure. Procured following UTS Procurement Procedure; assets remain UTS property and are entered in the project asset inventory. Total is 12% of the grant, within the 40% ceiling, Guideline Section 7(g). **Y2 (RM0):** No accessories required.	3,000.00	0.00	3,000.00
	TOTAL		**17,150.00**	**7,850.00**	**25,000.00**

VOT 23000 and VOT 28000 are not used. Virement may be exercised only once during the project and requires the approval of the Director, Centre for Research and Development. Supplier unit costs are planning figures pending current quotations, except the Sanger sequencing rate of RM25.00 per reaction, which is quoted.